RESUME

Chanhee Lee

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PROJECT INTERESTS

- **Embedded system HW/SW co-design, software platform development, architecture optimization, CXL-based memory systems, AI and on-device/edge LLM systems, and AI performance analysis**

SUMMARY

- **Embedded system, software platform, and AI systems engineer with 9+ years at Samsung Research and recent postdoctoral work on CXL-based memory systems and on-device LLMs**
- **Designed production AI inference platforms for Android/TVs and distributed software frameworks for IoT**
- **Interested in embedded system HW/SW codesign for AI and robotics, and LLM framework optimization**

PROFESSIONAL EXPERIENCE

- **University of Central Florida**, Orlando, FL, United States
  Postdoctoral Researcher, Department of Computer Science **Dec. 2024 - Present**
  • Designed memory-efficient LLM inference for CXL memory systems (Gem5 simulation) [gem5].
  • Extended Linux kernel for fault-tolerant non-volatile memory (NVM) in real-time edge systems [linux_pmo].
  • Leading 7 students to analyze on-device mlc-llm/llama.cpp performance and personalization with vector database.

- **Arizona State University**, Tempe, AZ, United States
  Visiting Scholar (Volunteer), School of Computing and Augmented Intelligence **Jan. 2024 - Dec. 2024**
  • Built Rust reactive engine in Lingua Franca (CAP-theorem guarantees) [lf-rust-rti].
  • Led on-device LLM personalization (Android + knowledge graphs) [EMSOFT WIP 2024] [WWW Short 2025].

- **Samsung Electronics Inc.**, Seoul, Republic of Korea
  Staff Engineer/Best Reviewer, Platform Team, Samsung Research **Aug. 2014 - Dec. 2024**
  • Presented platform team demo at Samsung Research annual showcase (**Top 3/10 major divisions**, ~2K researchers), on-device/remote inference with Qualcomm SoC GPU/NPU acceleration with SmartTrainer TV app
  • Led SmartFS filesystem development (fixed power-cut crashes for Samsung home appliance production boards).
  • Ported Oxigraph graph database to Android smartphone for edge AI workloads and developed REST APIs.
  • Developed OCF IoTivity D2D framework for ARTIK production boards (1st author, Springer LNCS ICIoT 2018) [RT-OCF].
  • Led two overseas research teams in the UK and Ukraine to develop GUI-based software performance analysis tools.

EDUCATION

- **Ph.D.** Electrical Engineering & Computer Science, Seoul National University 2014 (Advisor: Soonhoi Ha)
  Thesis: Fault-aware Task Mapping and Resource Management Techniques for Many-core Accelerators
- **M.S.** Computer Science, KAIST 2009 (Advisor: Doohwan Bae) | **B.S.** Computer Science, KAIST 2007

SKILLS

- **Languages:** Rust, C/C++, Python, Java, SystemC
- **Platform/Infra:** CXL/Gem5, Linux kernel, embedded systems, QEMU, NoC, Git, Docker, CI/CD, TDD, Agile Process
- **AI/ML:** llama-cpp/MLC-LLM, PyTorch/TensorFlow, RAG, Knowledge Graph, on-device ML performance optimization

PUBLICATIONS

- **8 conference** papers (EMSOFT, WWW) + **2 journal** papers (JSPS, TECS) | 250+ citations [Google Scholar]

AWARDS & LEADERSHIP

- **UCF Preeminent Postdoc Fellow (2025-26), Samsung Ph.D. Scholarship (2013)**
- **Invited Talk:** "Remote inference with IoT" (Hanyang University, South Korea 2023)
- **Reviewer:** ISCA/ASPLOS/HPCA/PPoPP/LCTES (2025) and ICS (2026), **ACM Member**